University of Glasgow

James Watt School of Engineering

Degrees of MEng, BEng, MSc and BSc in Engineering

ENG1064 Microelectronic Systems 1

Friday, 23 May 2025, Release time: 09:30-11:00

Exam duration: 90 minutes to complete exam plus 30 mins to upload submission

Attempt ALL questions.

TOTAL MARKS AVAILABLE
50

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

A calculator may be used. Show intermediate steps in calculations.

OVER

Q1 – Embedded Systems

Most of the electronic devices that we buy nowadays are based on microcontroller electronics and are built to continuously perform a particular task.

a) Briefly explain what an embedded system is. Detail the main functionalities of such a system and its differences from a personal computer.

[3]

b) Below, Figure Q1 shows an example of a modern smart vacuum. This is an embedded system with many of its functionalities based on microcontroller electronics.

   i. Detail the embedded functionalities of a conventional vacuum (not smart), i.e., what are the inputs, outputs and what information is being processed by a conventional vacuum's microcontroller. (Use a maximum of 200 words)

   [3]

   ii. Expand your previous answer to describe the embedded functionalities of the modern smart vacuum (e.g., phone connectivity), shown below. (Use a maximum of 300 words)

   [6]

   iii. You are an embedded systems developer working for a robotics manufacturer. Your company is developing the next generation of smart vacuums, and they are looking for a novel vacuum functionality that the company will incorporate in their next generation vacuums.
   Develop and compose a new, smart feature for a vacuum. This can be an imaginative feature. Make all the necessary links to embedded functionalities, i.e., what are the input/output signals, and information is being processed. (Use a maximum of 300 words).

   [3]

Figure Q1 Smart Vacuum

Subtotal [15]

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Q2 – Number Conversion

Digital electronics is based on the binary number system, where 0 and 1 are the only numbers used, and this is a number system with base-2 (2^N). Imagine that you are sent to a parallel universe where instead of a binary number system people used a new number system with their computers. This is called **pental** system and it is based on **pents** instead of our usual bits. The **pental** system is a **base-5** (5^N) number system and quins can have values of [0, 1, 2, 3, 4]. By extending your knowledge on the binary system, please answer the questions below.

a) For a 6-pent system, what is the maximum decimal number we can create?

[2]

b) Convert the pental number 11100110 to decimal.

[2]

c) Verify that the pental number 42301 corresponds to 0xB0A in the hexadecimal number system.

[3]

d) In this imaginary world with the pental number system, a microcontroller has address and data buses made of 8 parallel wires. Assuming each wire can carry one pent, what is the maximum number of memory locations it can have?

[1]

e) A 6-pent analogue to digital converter (ADC) in this parallel universe is connected to a reference voltage of 5V.

   i) Calculate the resolution in mV (milli-volts).

   [1]

   ii) Calculate the range in V (volts).

   [1]

Subtotal [10]

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Q3 – Pulse Width Modulation

Using your Nucleo Mbed, which has a reference voltage of 3.3V, you are trying to drive a motor which requires 10 Volts, at 1700rpm. However, you find that the motor does not operate as you initially expected.

a) What issues arise if you connect a motor directly to a Digital Output pin of your microcontroller?

[2]

b) Describe, in detail, why it would be more appropriate to use PWM when driving a motor.

[2]

c) Explain, using appropriate diagrams, how to connect the motor and MCU, such that the microcontroller is suitably protected from the inductive load of the motor.

[5]

d) On two different pins of your microcontroller, you decide to connect LEDs. You would like to connect a red LED to one output, which will emit light when the pin is digital high, and a blue LED to another output, which will emit light when the pin is digital low. The recommended operating current for these LEDs is 10 mA.

Sketch a diagram demonstrating how the two LEDs could be connected to the microcontroller. Considering the I-V characteristic curve of the two LEDs shown in Figure Q3 below, calculate the resistor value that should be used alongside each LED.

[6]

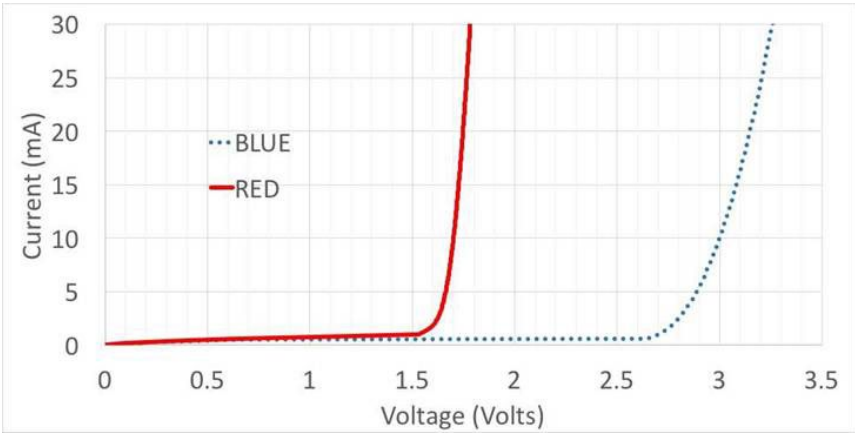

Figure Q3: Characteristic I-V curve of a red and a blue LED

Subtotal [15]

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Q4 – Communications: I2C

Working as an embedded systems developer at the robotics company, you are asked by your team leader to replace the smart vacuum's old analogue light sensor with a new digital sensor. According to the manufacturer, the **address** for this sensor is **0xA.** You would like to take a light reading from this sensor, and for that you **send the command 0x8**, while the I2C communication frequency is set at the recommended 100kHz.

a. Draw a diagram that illustrates the information sent through the Data and Clock wires. Give details on each segment of the communication and put appropriate labels on the Voltage and Time axes.

[8]

b. In this simplified system, where both addresses and data are 4 bits long, would it be possible to connect 20 devices using I2C? Justify your answer.

[2]

Subtotal [10]

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End of question paper